

Skillomate Production Deployment Migration

Version comparison and verification report | 18 September 2026 | Production EC2 43.205.137.167

CUTOVER	PM2	HEALTH	COURSES
SUCCESS	ONLINE	PASS	PASS

Executive outcome

The single production EC2 backend was migrated from a Mac/rsync source model to a GitHub-main source model. The live backend now runs the exact approved commit from the read-only GitHub repository, while production secrets remain on EC2. The original rsync application directories were preserved for emergency recovery.

Repository	Office50505/Edu-nex_Final
Deployed commit	260687cbd89b26ebbd9aefc9807fa86c4290b78d
Git checkout	/home/ubuntu/skillomate_repo
Backend	/home/ubuntu/skillomate_repo/edunex-b
Stable PM2 path	/home/ubuntu/skillomate_backend (symlink)
Runtime	skillomate_backend on port 3000

Current deployment flow

Approved merge to GitHub main -> one Mac command -> EC2 fetches origin/main -> exact commit reset -> locked dependency install -> existing PM2 process restart -> API verification -> success or automatic rollback.

Previous vs new deployment version

Area	Previous version	New version
Source of truth	Files present on a developer Mac.	GitHub origin/main at a known commit.
Transfer	rsync copied local application files to EC2.	EC2 fetches directly through a read-only deploy key; the Mac sends no source.
Reproducibility	npm install could resolve newer compatible versions.	npm ci --omit=dev uses the committed lockfile exactly.
Production state	The active directory was not a Git worktree.	A full monorepo checkout is retained at /home/ubuntu/skillomate_repo.
Runtime path	PM2 ran directly from the rsync directory.	The same stable PM2 path is a symlink to the repository backend subdirectory.
Secret handling	.env remained on EC2 but had permission 0644.	.env remains untracked on EC2 with permission 0600 and a secured backup.
Change protection	Local uncommitted files could be included in a deployment.	Dirty tracked or unexpected untracked production files stop deployment.
Concurrency	No deployment lock was defined.	flock rejects overlapping deploy attempts with exit code 75.
Verification	Process restart was the primary success signal.	PM2, port 3000, /api/health, and JSON shape of /api/courses must all pass.
Failure recovery	Manual recovery depended on the previous rsync workflow.	Future releases automatically reset to OLD_COMMIT, reinstall its dependencies, restart, and recheck.
Traceability	No standardized release record.	Timestamped logs record old/new commits, result, and rollback outcome without secrets.

Implementation delivered

Production tooling

- /home/ubuntu/bin/deploy-skillomate.sh - guarded production deployment with automatic Git rollback.
- /home/ubuntu/bin/test-deploy-skillomate-rollback.sh - isolated temporary-repository rollback simulation.
- /home/ubuntu/skillomate_deployments/ - protected timestamped deployment records.

Operator tooling

- /Users/kratik/deploy-skillomate-production.sh - the one-command Mac entry point.
- deployment/ - reviewable copies of the scripts and operational README in the local project.

Safety controls

- Dedicated GitHub SSH alias and repository-scoped read-only deploy key.
- Exact origin/main commit selection rather than a merge-based git pull.
- Pre-deploy checks for branch, remote, symlink target, .env, lockfile, and clean Git state.
- Fifteen health attempts at two-second intervals before a release is accepted.
- Non-zero exit on failure, including lock contention and recovered deployment failures.

Backup and recovery inventory

Artifact	Preserved location
Preparation copy	/home/ubuntu/skillomate_backend_rsync_backup_20260918-110521
Immediate pre-cutover copy	/home/ubuntu/skillomate_backend_pre_git_20260918-111148
Environment backup	/home/ubuntu/.skillomate_secrets/skillomate-production-20260918-110521.env
PM2 configuration	/home/ubuntu/.skillomate_secrets/skillomate-pm2-dump-20260918-110521.pm2

Testing and production verification

Test	Result	Evidence
Production/source drift	PASS	Live source matched known Git commit f5dc1c2 by checksum; only generated K6 reports were extra.
Candidate installation	PASS	141 locked production packages installed with npm ci.
Controlled cutover	PASS	Existing PM2 process restarted through the stable symlink.
Git state	PASS	HEAD and origin/main both equal 260687c; worktree is clean.
Runtime	PASS	PM2 online and port 3000 listening.
Health API	PASS	HTTP success with Skillomate API ok response.
Courses API	PASS	HTTP success, valid JSON, top-level list, one item at verification time.
Rollback simulation	PASS	Forced failure in an isolated temporary Git repository restored OLD_COMMIT.
Concurrency	PASS	A second deployment was rejected with exit code 75 while the lock was held.
Mac one-command path	PASS	Fetches GitHub and completed the already-current health path without restart.

Operator command

```
/Users/kratik/deploy-skillomate-production.sh
```

Scope intentionally unchanged

This version remains a controlled single-EC2 manual deployment. It does not add GitHub Actions, load balancing, Auto Scaling, additional EC2 instances, Docker, ECS, CodeDeploy, DNS changes, network changes, database changes, or application behavior changes.

Repository note

The deployment/ tooling copies are present as local project changes for review. The installed EC2 script and Mac helper are already active and verified; no source code is uploaded from the Mac during deployment.